

Soham Jagrit

Machine Learning Engineer

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Education

University of Texas at Arlington, Masters in Data Science

Aug 2024 - May 2026

Relevant courses: Machine Learning, Data Mining, Statistical Methods, Project Management

Arlington, TX

Ahmedabad University, Bachelors in Computer Science

Aug 2020 - May 2024

Relevant courses: AI, Big Data Analytics, Deep Learning, SQL

Ahmedabad, India

Skills

Frameworks And Libraries: PyTorch, TensorFlow, Scikit-learn, XGBoost, LangChain, Docling, Pandas, Numpy,

Languages: Python, C++, R, SQL

Other Tools: AWS (S3, Lambda, SageMaker, EC2), Docker, Git, GitHub Actions, ELK Stack, Pinecone

Data Science: Hypothesis Testing, Large Language Models, Model Training, Model Evaluation and Selection, Model Deployment, Context Engineering, LLM Fine Tuning, Model Quantization

Visualization Tools: Power BI, Tableau, Kibana, Matplotlib

Experience

Delta Air Lines

Jan 2026 - May 2026

Graduate Data Science Intern

Atlanta, GA

- Built a GPU-accelerated inference pipeline using Docling's ML-based layout and table detection models on AWS SageMaker (ml.g5.xlarge), achieving 12x throughput improvement processing 2,000+ PDFs in 5 hours vs 60+ hours
- Engineered a conditional inference system with an OCR toggle that reduced latency and unnecessary model forward passes by routing non-scanned PDFs to a standard backend, resolving OOM errors via post-batch memory cleanup
- Configured end-to-end SageMaker model training pipelines with PyTorch-backed AWS DLCs, entry point scripts, and S3 manifest files enabling batch-aware, resume-supported distributed document processing on GPU clusters
- Designed a human-in-the-loop model evaluation pipelines across coherence, factual accuracy, and usability metrics, feeding structured human scores into an LLM-as-judge prompt to drive iterative prompt optimization for AI generated summaries

University of Texas at Arlington

Mar 2025 - Dec 2025

Student Data Analyst

Arlington, TX

- Built a unified Power BI reporting system for UTA Admissions, migrating manual Excel tracking into a single real-time dashboard and saving 80+ hours of manual data management per month
- Delivered automated reporting by surfacing student engagement trends across all 3 channels- calls, emails, walk-ins, providing real time insights to the upper management

Softvan Pvt. Ltd.

Jan 2024 - Jun 2024

Data Science Intern

Ahmedabad, India

- Built a two-stage NLP pipeline on 50,000+ customer reviews, fine-tuned BERT for 3-class sentiment classification and used predicted polarities with demographic features to impute sentiment for customers who skipped open-ended feedback
- Engineered the full text preprocessing pipeline using NLTK, TF-IDF, and word embeddings - handling schema validation, missing value imputation, and structured data formatting for downstream machine learning model training

Travel Designer Group Pvt. Ltd.

Jun 2023 - Sept 2023

Data Engineering Intern

Ahmedabad, India

- Configured an ELK stack pipeline (Elasticsearch, Logstash, Kibana) that ingested and indexed 1M+ search records from a REST-based booking engine API, enabling real-time and historical insights into regional demand trends
- Designed the Elasticsearch index schema and built Kibana dashboards surfacing geographic search patterns, giving business stakeholders visibility into regional demand trends across searches

Projects

NutriBot

Python, Flask, LangChain, Hugging Face, Pinecone, Docker, AWS

- Built a RAG-based nutrition Q&A chatbot using LLaMA (Groq API), embedding 12,000+ text chunks from mixed nutrition sources into Pinecone vector storage to deliver grounded answers to user queries
- Used LangChain for retrieval of chunks and augmented response generation, deployed NutriBot on AWS EC2 instance with a Docker-containerized backend and GitHub Actions

Loan Defaulters Prediction

Python, Flask, Scikitlearn, Docker, AWS

- Built an end-to-end MLOps pipeline for predicting loan defaulters covering data ingestion, feature transformation, model training, and real-time prediction using Random Forest, achieving 93% F1 score.
- Containerized the application with Docker, automated model updates with CI/CD and deployed on AWS EC2 - establishing a production-ready machine learning delivery workflow